

Josh Talks AI — Transcriber Quality Check

Part I: Identifying Low-Quality Transcribers from Task-Level Data

1. Dataset overview

- 51,783 completed transcription tasks across 48,875 unique users.
- 48,865 users completed exactly one task; 10 users completed between 86 and 609 tasks each. Because a “pattern” requires repeated behaviour, the 10 repeat contributors are the most reliable group to study — one-off tasks from single-task users are used only to build the baseline (“normal”) distribution for comparison.
- Fields used: `is_edited`, `duration`, `time_taken_by_user`, `segment_character_per_second`, `whisper_text`, `user_text`.

2. Method

For every user, three rates were computed across their tasks and compared against the platform-wide baseline (the 48,865 single-task users). A user is a candidate for review when a rate sits well above that baseline — not on one task, but consistently across their history.

3. Warning sign 1 — Finishing faster than the audio itself

Metric: $\text{time_ratio} = \text{time_taken_by_user} \div \text{duration}$

What it tells us: If $\text{time_ratio} < 1$, the transcriber submitted their answer in less time than the clip takes to play. Since `time_taken_by_user` includes listening to the audio, this is only possible if they scrubbed through, skipped playback, or guessed without listening at all.

How to measure: For each user, % of their tasks where $\text{time_ratio} < 1$, computed only on tasks with a valid (non-missing) `time_taken_by_user` value.

Red flag threshold: Baseline rate is 10.5%. All 10 repeat contributors sit between 24.2% and 33.9%, roughly 2.5–3x the baseline. Rather than a fixed number, the threshold is defined relative to baseline — flag when a user's rate exceeds ~2x the platform baseline, sustained over enough tasks to rule out chance.

False alarms: Initially suspected that short clips (many clips in this dataset are under 1 second) could cause innocent $\text{time_ratio} < 1$ flags — e.g., a transcriber who already previewed a very short clip finishing “before it ends” without having skipped anything. This was checked directly and ruled out: among all flagged rows, the minimum clip duration was 0.87 seconds and the average was 9.6 seconds — actually *longer* than the dataset average (5.5s). The flagged cases are concentrated on longer clips, not short ones, which makes the signal harder to explain innocently, not easier. One remaining data-quality caveat: 180 rows (0.3% of the dataset) have a missing `time_taken_by_user` value; these should be explicitly excluded from the calculation rather than left in, though confirmed they don't fall on any of the 10 flagged users and so don't change the findings above.

Conclusion: 9.960% users are with more than 21.1% of tasks having $\text{time_ratio} < 1$.

4. Warning sign 2 — Implausible typing speed

Metric: `segment_character_per_second` (non-space characters typed ÷ time taken)

What it tells us: Human transcription speed clusters tightly — the platform-wide median is 10.4 characters/second. A user who is repeatedly far above that isn't typing character-by-character from listening; they are likely pasting boilerplate text, mashing keys to pass the task quickly, or the timer isn't capturing real effort.

How to measure: % of a user's tasks (restricted to clips longer than 1 second, to avoid divide-by-tiny-time noise) where `segment_character_per_second` > 30 (roughly the 95th percentile platform-wide).

Red flag threshold: Baseline rate is 5.5%. User 688393 sits at 15.1% — about 3x baseline — the clearest outlier on this signal.

False alarms: Very short replies (“हां”, “जी”) divided by a tiny `time_taken_by_user` can spike this ratio purely from small-number arithmetic, even for an honest, fast transcriber. Restricting the calculation to clips above a minimum duration (e.g. > 1 second) removes most of this noise.

Conclusion: Total 20 Users are with more than 10% of tasks having 'suspiciously fast typing' (>30 chars/sec):

6. Summary comparison table

Shaded cells mark the standout figure discussed for that user above.

User ID	Tasks done	Edit rate	% time_ratio < 1	% high CPS (>30)	% short text (≤3 chars)	% identical to Whisper
580062	609	87.2%	28.2%	7.9%	11.0%	12.8%
393886	509	96.9%	25.9%	5.3%	19.8%	3.1%
681509	358	99.2%	30.2%	5.6%	3.9%	0.8%
410566	349	91.7%	30.1%	7.2%	8.0%	8.3%
222099	333	98.8%	28.2%	10.5%	3.3%	1.2%
311760	184	100.0%	28.3%	4.3%	3.3%	0.0%
467270	176	97.2%	27.3%	5.7%	7.4%	2.8%
632098	165	57.6%	33.9%	3.0%	17.0%	42.4%
326611	149	100.0%	24.2%	5.4%	7.4%	0.0%
688393	86	83.7%	31.4%	15.1%	3.5%	16.3%

Baseline (48,865 single-task users): edit rate 82.9% · % time_ratio<1 = 10.5% · % high CPS = 5.5% · % short text = 6.3% · % identical to Whisper = 30.8%

7. Takeaway

- No single metric is sufficient grounds for action on its own — each has a plausible innocent explanation. However, considered together, the data supports two levels of concern.
- At the platform level, all 10 repeat contributors show a `time_ratio < 1` rate of 24.2–33.9%, consistently 2.5–3x the 10.5% baseline. This was specifically tested against the hypothesis that short clips could explain it away, and that hypothesis did not hold — the pattern persists even after accounting for clip length. This indicates the behaviour is systemic rather than isolated to a few individuals, and warrants monitoring across the broader user base, not just the repeat contributors examined here.
- At the individual level, one user — 632098 — stands out as a distinct case requiring closer review. This account combines a below-baseline edit rate (57.6% vs. 82.9%) with an above-baseline identical-to-Whisper rate (42.4% vs. 30.8%). The co-occurrence of these two signals is materially stronger evidence of low-effort work than either figure alone, since it points to a consistent behaviour (accepting the AI draft without correction) rather than a statistical artifact.
- Taken together, these findings support a two-tier response: broader platform monitoring for the time-ratio pattern, and a targeted, prioritized review of account 632098 based on the compounded signal it shows.

Part II: Detection System for Problematic Transcribers

1. Validated metrics used in blocking logic

Tier	Metric	Status	Why
Blocking-eligible	<code>time_ratio < 1</code> rate	Validated	Held up after testing and ruling out the short-clip explanation.
Blocking-eligible	Low edit rate + high identical-to-Whisper (compound)	Validated	Meaningful only when both co-occur; isolates rubber-stamping from easy-audio cases.

2. Blocking logic

Flag a transcriber's account for suspension review if, over their trailing 50 tasks:

Rule 1 — Rushing: `time_ratio < 1` rate > 2x platform baseline, on ≥30 of 50 valid-timing tasks.

Rule 2 — AI rubber-stamping: edit rate < 0.5x baseline AND identical-to-Whisper rate (`whisper_text > 10` chars only) > 1.5x baseline — both required.

Both rules require co-occurrence with themselves across the window — a single task never triggers a flag.

Either rule → manual review, not auto-suspension. Repeat trip after review → suspend.

3. Explicitly excluded from blocking logic

segment_character_per_second (typing speed) is not used to gate any account action. It is retained as a monitoring/research metric only — see the rejection table below for why.

4. Candidates tested and rejected

Four additional signals were tested against this dataset using the same method as the validated metrics above (compare repeat contributors to platform baseline, then attempt to explain the pattern away). Only the two in Section 1 survived.

Candidate signal	Result	Why rejected
Typing speed (segment_character_per_second > 30)	Rejected	Baseline collapsed from 5.5% to 0.06% once short clips ($\leq 1s$) were excluded — almost entirely a small-number-arithmetic artifact, not real behaviour.
Duplicated / copy-pasted text across clips	Rejected	Initial 13–28% rates traced to legitimate recurring phrases (“ठीक है”, the platform's own “[blank]” convention for silent audio). At a strict ≥ 8 -word-sentence definition, the rate was 0% for all 10 repeat contributors.
Trivial edits to game the is_edited flag	Inconclusive — monitor only	Baseline itself rises from 27% to 42% as text length increases, and one repeat contributor scored below baseline at every cutoff tested. Not a clean, consistent pattern — not used for blocking.
Under-transcription relative to clip length	Rejected	Baseline rate 5.2%; 6 of 10 repeat contributors were at exactly 0%, and the highest was still below baseline. No separation found.

5. Engineering build list

- Scheduled job to recompute platform-wide baselines (e.g., weekly).
- Per-user rolling 50-task calculation, refreshed after each submission.
- Review queue (not block queue) with flagged tasks/metrics attached.
- Escalation record per user to distinguish first flag vs. repeat.

6. Summary

Block only on sustained Rule 1 or Rule 2 behaviour — never on typing speed alone, never on a single incident, never without human review. Four other candidate signals were tested and did not meet this bar.